Avatar Assisted AI Health Assessment Across Five Screening Domains

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Abstract

This paper presents a unified health assessment framework that connects structured intake, an avatar assisted interview, risk categorization, and reviewable reporting. Cardiovascular, neurological, full health, respiratory, and psychometric modules share one workflow. An OpenAI language model supports conversation and summarization; HeyGen supplies avatar presentation, Wispr Flow supplies dictation, and MongoDB stores assessment records. A critical literature survey distinguishes interaction quality, clinical evidence, and prediction performance. An educational experiment compares three classifiers using 5,000 synthetic records, equally allocated across modules. Logistic regression achieves 93.60% accuracy and 93.78% macro F1 on 1,000 held-out records. These results measure agreement with artificial priority labels. The contribution is an integrated project design and a transparent evaluation example, with traceable evidence and human review treated as core requirements.

Keywords: conversational AI; health assessment; virtual avatar; risk categorization; synthetic evaluation.

1 Introduction

Health assessment begins with collecting information, but its usefulness depends on what happens next. A symptom description may be incomplete, a lifestyle response may need context, and an uploaded report may contain details that the user has not mentioned. A reviewer needs these pieces to form a coherent history. When intake forms, interviews, and report preparation operate separately, important context can be lost between stages. An effective digital assessment workflow should therefore support clarification, preserve the source of each relevant statement, and present the resulting information in a form that can be checked.

Conversational AI offers a way to guide this process through questions that respond to what a person has already said. However, a natural conversation is only one part of an assessment system. Reviews of health conversational agents describe promising applications while identifying limited or inconsistent evidence of clinical effectiveness and safety [1, 2]. This distinction motivates the present project: the goal is to connect conversation to structured information and accountable review, with each stage assigned a clear responsibility and an appropriate evaluation target.

The proposed system brings cardiovascular, neurological, full health, respiratory, and psychometric assessment into one platform. These modules share account access, consent, personal details, interview controls, and report navigation. Their questions and interpretation requirements remain domain specific. For example, a respiratory interview collects reported breathing concerns, whereas a psychometric interview concerns self-reported psychological or functional information. A common interface can make the workflow consistent without implying that the modules measure the same clinical construct or can use one interchangeable diagnostic rule.

The technical design separates the interaction layer from the prediction and reporting services. An avatar presents questions, a language model supports clarification and draft summarization, and a structured classifier estimates review priority from defined inputs. Keeping these functions separate allows a reviewer to distinguish a user statement from an extracted fact or a predicted category. It also makes failures easier to locate: a transcription error, a missed follow-up question, and an incorrect priority assignment require different remedies. The final report should retain uncertainty and missing information instead of presenting every generated statement as an established fact.

The project is intended to support health risk categorization, medical history summarization, symptom flagging for professional review, and authorized information gathering for insurance underwriting. Its contribution is an integrated workflow, a modular architecture covering all five assessment types, and a transparent example of how to compare prediction methods. The numerical study uses equally represented synthetic records and evaluates agreement with artificial priority labels. It demonstrates the analysis procedure while leaving clinical usefulness, report quality, and operational performance to separate evaluation. The following sections examine related work, describe the system and study methods, present the results, and discuss their practical limits.

2 Literature review and detailed survey

This is a focused narrative survey, rather than a systematic review or meta-analysis. Sources were selected for their relevance to conversational assessment, embodied interaction, domain measurement, prediction evaluation, and responsible implementation. Different study types are compared by the evidence they provide; their reported scores are not pooled.

Conversational systems and personalization

Laranjo et al. [1] reviewed health agents accepting unconstrained natural language and found that efficacy and safety were rarely evaluated. Milne-Ives et al. [2] reported promising usability and effectiveness findings but insufficiently strong evidence for broad conclusions. Kocaballi et al. [3] examined personalization and found little evaluation isolating its effect on health outcomes. Together, these reviews support adaptive questioning while showing why user satisfaction, successful conversation, and improved health should remain separate evaluation targets.

Avatars and relational interaction

Bickmore and Picard [4] studied a relational agent in a month-long exercise intervention with 101 users and observed stronger trust and willingness to continue interaction than with a task-oriented agent. Provoost et al. [5] reviewed embodied agents in clinical psychology and found heterogeneous applications and sparse effectiveness evidence. These findings motivate a clearly disclosed avatar interface; they do not establish that visual realism improves the accuracy of an assessment or its resulting report.

Medical language models and dialogue

Singhal et al. [6] introduced MultiMedQA and evaluated medical answers along factuality, reasoning, harm, and bias dimensions. Ayers et al. [7] compared physician and chatbot replies to questions from a public forum using evaluator ratings of quality and empathy; that design did not establish clinical outcomes. Tu et al. [8] evaluated AMIE in text consultations involving 159 simulated case scenarios and 20 primary care physicians. Its structured dialogue evaluation is more directly relevant to history-taking, although simulated patients and text chat limit transfer to routine care. This project therefore separates the language layer from the risk classifier and treats both as requiring their own evidence.

Evidence across the five assessment domains

Clinical assessment requires a definition of what is being measured. Table 1 gives each module an established reference point and an explicit boundary. These sources guide the evaluation design; their instruments and algorithms were not applied to the synthetic records. A common interface can organize five modules, but a shared low, moderate, or high label cannot make their clinical meanings interchangeable.

Table 1. Domain literature and implications for assessment design.

Module and source	Established contribution	Implication and boundary
Cardiovascular D'Agostino et al. [9]	Framingham models estimate general cardiovascular risk from measured risk factors.	Specify predictors, outcome and horizon; interview priority is not a cardiovascular event probability.
Neurological Nasreddine et al. [10]	MoCA evaluates a defined screening instrument for mild cognitive impairment.	Use a validated administration and scoring protocol; a symptom interview is not a cognitive test.
Full health WHO ICF [11]	A framework describes functioning, activity, participation and environmental context.	Preserve physical, functional and contextual history; the framework is not a universal diagnostic score.
Respiratory Graham et al. [12]	ATS/ERS standards define quality requirements for spirometric measurement.	Distinguish reported breathing symptoms from objectively measured lung function.
Psychometric Kroenke et al. [13]	The PHQ-9 evaluates a specific measure of depression severity.	Use an appropriate validated instrument when needed; it does not validate a generic personality assessment.

Prediction quality calibration and fairness

Steyerberg et al. [14] distinguish overall performance, discrimination, calibration, and decision usefulness. Van Calster et al. [15] explain why useful ranking can coexist with unreliable absolute risk estimates. Accuracy and macro F1 in this paper consequently describe classification agreement only. Obermeyer et al. [16] show how health expenditure used as a proxy for need can produce racial bias. This is particularly relevant to underwriting:

convenient administrative labels may reflect access or cost rather than health. Ghassemi et al. [17] caution that explanations alone cannot establish the quality of an individual prediction. Traceability and external evaluation must accompany any explanation.

Reporting and clinical evaluation

TRIPOD+AI specifies transparent reporting for prediction studies [18]. DECIDE-AI addresses early clinical evaluation and human factors [19], whereas CONSORT-AI extends reporting for trials of AI interventions [20]. These guidelines address different stages of evidence. The present synthetic experiment can illustrate prediction reporting; it cannot substitute for an operational study or randomized trial. Table 2 links the principal literature strands to concrete design choices.

Table 2. Comparison of clinical AI evidence and the remaining research gap.

Evidence strand	What it establishes	Remaining gap and design response
Medical LLMs [6, 7, 8]	Knowledge, response ratings and simulated consultation quality.	Benchmarks differ from operation; audit extraction fidelity and report completeness.
Domain measures [9, 10, 11, 12, 13]	Defined clinical constructs, protocols or measurement frameworks.	Shared navigation does not validate shared scoring; define outcomes for each module.
Prediction and bias [14, 15, 16, 17]	Complementary performance measures and failure mechanisms.	Assess calibration, proxy labels, subgroup errors and external transportability.
Evaluation guidance [18, 19, 20]	Reporting requirements across prediction studies and clinical evaluation.	Progress from transparent offline results to documented live and prospective evaluation.

3 System workflow and architecture

The design connects the literature through a shared workflow, traceable evidence, and standard statistical baselines [21]. The user signs in, selects an assessment, chooses a language, and gives consent. Shared intake records profile details, height, weight, smoking, exercise frequency, and optional medical documents. Each module supplies its own interview topics. Figure 1 shows the common path; the five branches have equal status.

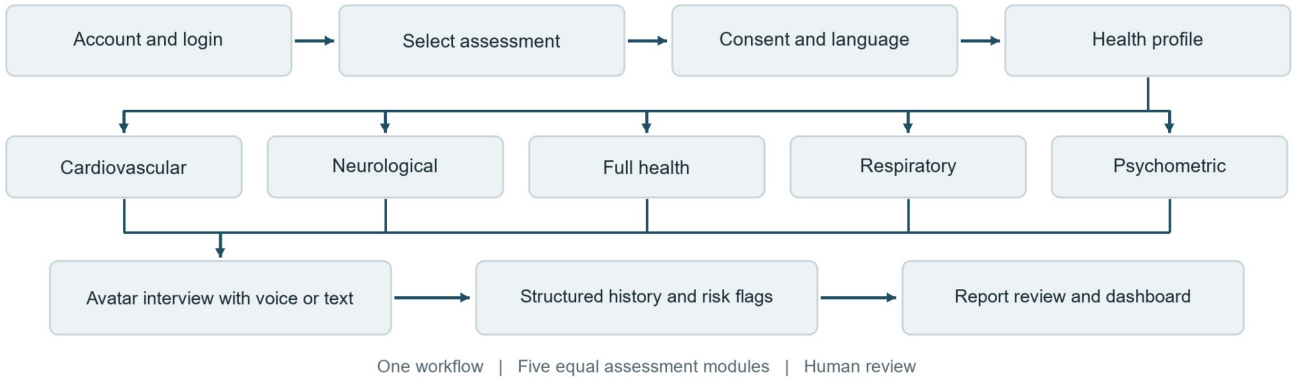


Figure 1. Common user journey across five assessment modules. Reports preserve confirmed history, missing information, risk flags and items for professional review.

The session asks one clear question at a time and resolves contradictory answers. Typed input provides an alternative to dictation. Users control camera access and can correct factual details before sharing. Report generation has explicit pending, available, and failed states so that reaching the review screen does not imply a report is ready.

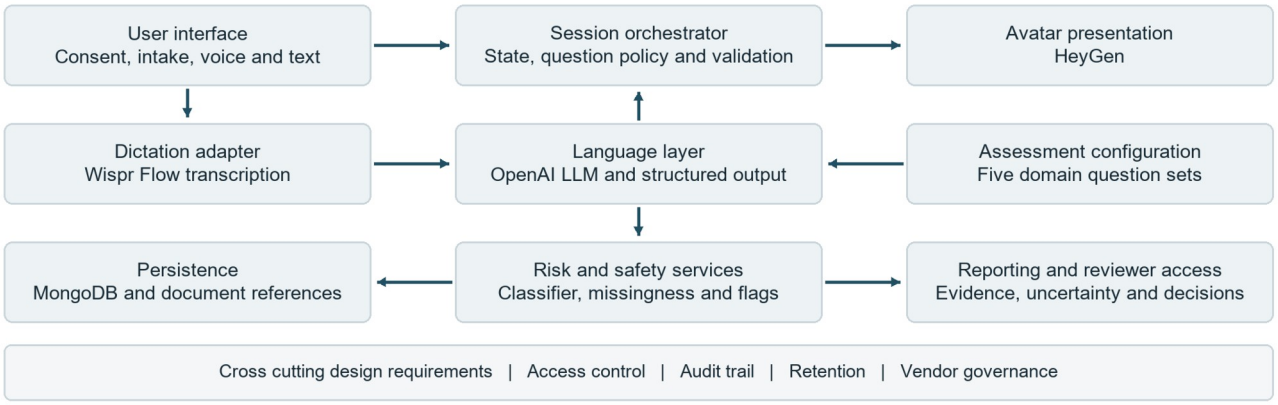


Figure 2. Proposed component responsibilities based on the reported stack. The diagram describes a design, rather than a verified deployment or security audit.

The OpenAI language layer handles question wording, clarification, extraction, and draft summaries. Structured Outputs can enforce a schema, but content still needs validation [22]. HeyGen presents generated speech through the avatar [23]. Wispr Flow supplies dictation; a proposed adapter passes confirmed text into the session, keeping transcription separate from question selection [24]. A production configuration should pin model versions and prompts.

MongoDB stores consent, interview turns, structured facts, prediction outputs, and report metadata in separate logical collections. Every derived fact should retain its source and timestamp. Uploaded files remain optional evidence; automated extraction would require its own evaluation. The prediction service consumes structured inputs, while the report distinguishes user statements from model interpretations. These boundaries support independent testing and accountable reviewer access.

4 Prediction methods and synthetic study design

The experiment evaluates low, moderate, and high review priority using 5,000 independent synthetic records, equally allocated across the five modules. Within each module, stratified sampling assigns 700 records to training, 100 to validation, and 200 to a held-out test set. No individual crosses splits. The three classifiers follow standard statistical learning formulations [21].

Table 3. Dataset splits and reference label counts across all 1,000 records per module.

Module	Train	Valid	Test	Low	Mod	High
Cardiovascular	700	100	200	302	398	300
Neurological	700	100	200	299	396	305
Full health	700	100	200	301	394	305
Respiratory	700	100	200	321	391	288
Psychometric	700	100	200	275	414	311
Total	3500	500	1000	1498	1993	1509

Data generation and artificial reference labels

Age is uniform from 18 to 80 years. Height follows a normal distribution with mean 168 cm and SD 9 cm, clipped to 140–198 cm. BMI follows a normal distribution with mean 26 and SD 4.5, clipped to 16–43; weight is derived from height and BMI. Smoking is Bernoulli with probability 0.24; exercise is uniform on 0–3. The generated mean age is 48.9 years (SD 18.2), mean BMI is 25.9 (SD 4.5), and 1,236 records report smoking (24.72%).

Independent ordinal descriptors represent symptom burden b (0–4), history burden h (0–3), and functional impact f (0–3), each sampled uniformly. Let s denote smoking, e exercise, and I an indicator. These descriptors are illustrative and are not the clinical instruments reviewed in Section 2.

$$\text{BMI} = \text{weight (kg)} / [\text{height (m)}]^2 \quad (1)$$

$$S = 1.20b + 0.90h + 0.80f + 0.40sI(b \geq 2) + 0.25I(e = 0) + \varepsilon \quad (2)$$

Noise ε is normal with mean 0 and SD 0.22. Labels are low for $S < 3.8$, moderate for $3.8 \leq S < 6.4$, and high otherwise. Thresholds are fixed before fitting and the random seed is 42. All modules use the same artificial rule.

Consequently, the labels concern a reproducible priority task and do not represent a diagnosis, event probability, or clinically calibrated risk horizon.

Model development and selection

All candidates use age, BMI, smoking, exercise, the three burden descriptors, two indicator interactions, and five one-hot module fields. Only training data determine scaling means and standard deviations. Multinomial logistic regression minimizes cross-entropy with L2 regularization ($\lambda = 0.0001$; intercept unpenalized). Adam uses 3,500 full-batch updates, learning rate 0.03, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\varepsilon = 10^{-8}$. Gaussian naive Bayes uses class-specific means, variances with 10^{-6} smoothing, and empirical priors. Nearest neighbours uses $k = 15$, Euclidean distance, and uniform voting. Settings are fixed in advance; validation macro F1 selects the final model. NumPy 2.3.5 performs the calculations.

The held-out set is used only after model selection. Precision, recall, and F1 are calculated per class and averaged without prevalence weighting. Wilson intervals quantify accuracy uncertainty; a 2,000-resample percentile bootstrap of fixed test pairs quantifies macro F1 uncertainty (seed 2026). Neither interval includes uncertainty from clinical deployment or repeated development.

5 Results of the synthetic evaluation

Validation macro F1 was 94.17% for logistic regression, 76.16% for Gaussian naive Bayes, and 79.36% for nearest neighbours. Logistic regression was selected before holdout inspection. Table 4 compares the fixed models on the same 1,000 test records.

Table 4. Holdout results against synthetic labels. All values are percentages.

Classifier	Accuracy	Macro precision	Macro recall	Macro F1
Logistic regression	93.60	93.87	93.70	93.78
Gaussian naive Bayes	77.60	78.21	78.88	78.10
Nearest neighbours	80.70	82.20	80.65	81.25

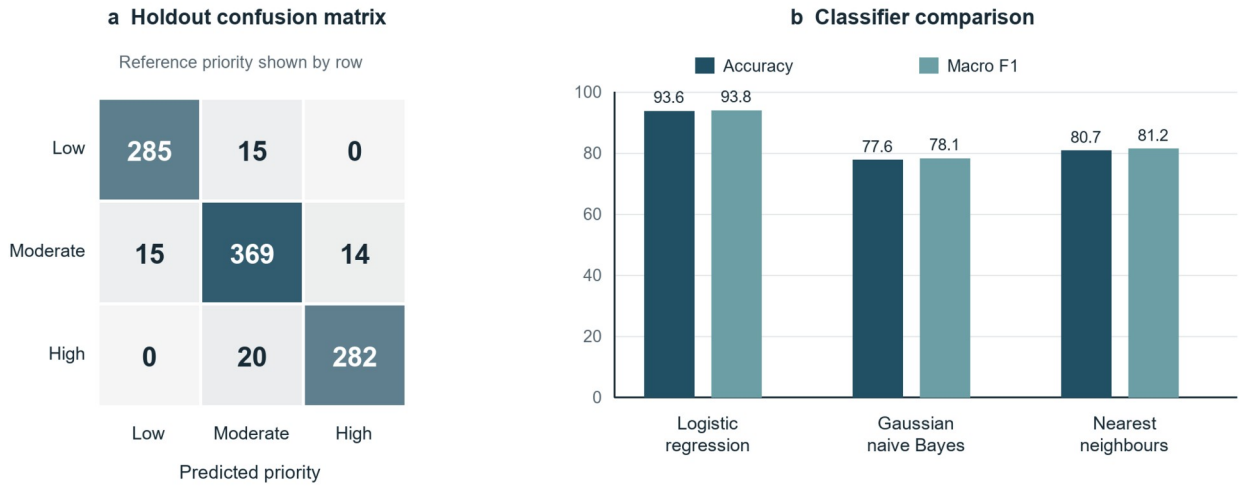


Figure 3. Logistic regression confusion matrix and comparison of classifiers. Reference class totals are 300 low, 398 moderate and 302 high.

The selected model classified 936 records correctly: accuracy 93.60% (95% Wilson CI 91.91–94.96%) and macro F1 93.78% (95% bootstrap CI 92.33–95.16%). It assigned 20 of 302 high-priority records to moderate priority, giving high-priority recall of 93.38%. No direct low-to-high or high-to-low errors occurred in this sample. A majority-class predictor would achieve 39.80% accuracy.

$$\text{Precision}_k = TP_k / (TP_k + FP_k) \quad \text{Recall}_k = TP_k / (TP_k + FN_k) \tag{3}$$

$$F1_k = 2\,\text{Precision}_k\,\text{Recall}_k / (\text{Precision}_k + \text{Recall}_k) \quad \text{Macro F1} = \Sigma_k F1_k / 3 \tag{4}$$

Table 5. Equal module representation in the selected model holdout results.

Assessment module	n	Accuracy %	95% Wilson CI %	Macro F1 %
Cardiovascular	200	94.0	89.81–96.53	94.17
Neurological	200	91.5	86.81–94.63	91.74
Full health	200	95.5	91.67–97.61	95.63
Respiratory	200	94.5	90.42–96.90	94.63
Psychometric	200	92.5	88.00–95.40	92.74

The overlapping intervals and common generator do not establish clinical differences between modules. Strong logistic performance is consistent with the additive label rule and supplied interactions. These results demonstrate recovery of an artificial relationship; speech accuracy, report quality, calibration, usability, and clinical outcomes were not measured.

6 Discussion and responsible implementation

Interpretation of the integrated design

The main project contribution is the connection between intake, adaptive questioning, structured risk classification, and report review. The literature indicates why those stages should be evaluated separately [1, 2, 6, 8]. A fluent interview can omit a critical detail, and a well-performing classifier can receive an incorrectly extracted input. Reports should therefore distinguish confirmed statements, unresolved answers, evidence from documents, and model interpretations. The same requirements apply across all five modules.

A low predicted priority must not suppress a separately defined urgent symptom pathway. Incomplete or contradictory histories should trigger clarification or professional review. Insurance staff should receive relevant, authorized information and retain responsibility for decisions. The proxy-label problem identified by Obermeyer et al. [16] means that cost, claims history, and access to care require particular scrutiny before they are treated as indicators of health need.

Privacy and HIPAA related design requirements

HIPAA applicability depends on the organization, information flows, and service relationships. HHS describes administrative, physical, and technical safeguards, including risk analysis and access management [25]. The project treats compliance as a development objective; it has not been verified. Wispr documentation requires appropriate settings and a signed Business Associate Agreement for PHI dictation [24]. Vendor capabilities do not establish that an individual deployment is compliant.

Table 6. Proposed operational controls and the evidence needed to assess them.

Area	Proposed control	Evidence required
Access and storage	Record ownership, role-based access, protected transport and storage	Configuration review, access tests and key management records
Consent and retention	Purpose-specific sharing, retention limits and deletion workflow	Consent versions, retention policy and deletion tests
Traceability	Source-linked facts and restricted audit logs	Log review, incident procedures and recovery tests
External services	Permitted data flows and applicable vendor agreements	Executed agreements and verified service settings

Limitations and an evaluation roadmap

The artificial labels are derived from the same structured fields the models receive, so strong performance is expected when a model approximates that rule. Independent feature sampling omits real correlations, missingness, measurement errors, and changes over time. Balanced module counts support comparison within this example but do not describe actual demand or demonstrate fairness. No patient records, clinician labels, participant recruitment, clinical follow-up, security audit, or external evaluation were used.

Future work should define outcomes independently for each module and test on representative, clinician-reviewed records. Prediction studies should report calibration, discrimination, uncertainty, and subgroup errors [14, 15, 18]. Dialogue audits should measure extraction fidelity, unsupported statements, and missing critical details. An avatar

versus text comparison could test whether embodiment adds value [4, 5]. Early live evaluations should examine clinician interaction and workflow failures using DECIDE-AI principles [19]; any later trial should use appropriate CONSORT-AI reporting [20].

7 Conclusion

A shared assessment architecture can serve five modules while preserving domain-specific interview content and accountable reporting. The synthetic experiment illustrates transparent model selection and error analysis. The literature survey shows that conversational quality, clinical measurement, prediction performance, and operational governance require distinct evidence before deployment claims can be justified.

Study declarations: no human participants or patient data were used. No ethics approval, external funding, clinical registration, or regulatory authorization is claimed. Vendor names identify component roles and do not imply endorsement.

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